BIOLOGY ALL IN ONE (BAIO) LEARNING SYSTEM

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BIOTECHNOLOGY: PRINCIPLES AND PROCESSES

Quick Revision & Concept Summary Notes

1. INTRODUCTION & CORE PRINCIPLES

- **EFB Definition:** "The integration of natural science and organisms, cells, parts thereof and molecular analogues for products and services."
- **Two Core Techniques of Modern Biotechnology:**
  - **Genetic Engineering:** Techniques to alter chemistry of genetic material (DNA/RNA) to introduce into host organisms and change phenotype.
  - **Bioprocess Engineering / Aseptic Techniques:** Maintenance of sterile environment to grow only desired microbes in large quantities for products (vaccines, enzymes, antibiotics).
- **Three Basic Steps in GMO creation:** (1) Identification of DNA with desirable genes → (2) Introduction into host → (3) Maintenance & transfer to progeny.

2. TOOLS OF RECOMBINANT DNA TECHNOLOGY (RDT)

A. Restriction Enzymes ("Molecular Scissors")

- Belong to class **Nucleases**:
  - **Exonucleases:** Remove nucleotides from the ends of DNA.
  - **Endonucleases:** Cut DNA at specific positions within the DNA. Over 900 isolated from 230 strains.
- Recognize specific **Palindromic Recognition Sequences** (reads same in $5' \rightarrow 3'$ and $3' \rightarrow 5'$ direction).
  Example (EcoRI site):
  $5' - G \mid A\,A\,T\,T\,C - 3'$
  $3' - C\,T\,T\,A\,A \mid G - 5'$
- Cuts slightly away from center of palindrome, producing single-stranded overhanging ends called **Sticky Ends** (facilitate ligation via H-bonding).
- **Naming Scheme (e.g., EcoRI):** 1st letter = Genus (*Escherichia*), 2nd & 3rd letters = Species (*coli*), 4th letter = Strain (*R*), Roman numeral = Order of discovery (*I*).

B. Gel Electrophoresis (DNA Separation)

- Negatively charged DNA fragments move towards the **Anode (+ pole)** through an **Agarose gel** matrix.
- Sieving effect: Smaller fragments travel faster and farther.
- **Visualization:** Stained with **Ethidium Bromide (EtBr)** and exposed to **UV light** → Bright Orange bands.
- **Elution:** Cutting out DNA bands from agarose gel and extracting them.

C. Cloning Vectors

- Plasmids and Bacteriophages. Essential features:
  - **Origin of Replication (ori):** Sequence where replication starts; controls copy number.
  - **Selectable Markers:** Identify and select transformants/recombinants while eliminating non-transformants. Commonly antibiotic resistance genes (amp^R, tet^R).
  - **Cloning / Restriction Sites:** Unique target sites for restriction enzymes.
- **Vector pBR322 Features:**
  - Restriction sites: *BamHI, SalI, PvuI, PstI, ClaI, HindIII, EcoRI, PvuII.*
  - Antibiotic genes: amp^R (has *PstI, PvuI*) and tet^R (has *BamHI, SalI*).
  - *rop* gene: Codes for proteins involved in plasmid replication (contains *PvuII* site).
- **Insertional Inactivation:** Insert of foreign DNA into a functional gene inactivates it.
  - *Selection using β-galactosidase gene:* Recombinants lose enzyme activity → form **white colonies** on chromogenic substrate. Non-recombinants produce active enzyme → **blue colonies**.
- **Vectors for Plants/Animals:**
  - *Ti Plasmid* of *Agrobacterium tumefaciens* (disarmed tumor-inducing plasmid for plants).
  - *Retroviruses* (disarmed oncogenic viruses for animal cells).

D. Methods of Gene Transfer into Competent Host

- **Chemical Transformation:** Divalent cations (Ca^{2+}) + Ice incubation → Heat shock (42°C) → Ice.
- **Microinjection:** Recombinant DNA directly injected into nucleus of animal cell.
- **Biolistics / Gene Gun:** High-velocity microparticles of gold or tungsten coated with DNA shot into plant cells.
- **Disarmed Vector Infection:** Pathogen vector modified to infect cells without causing disease.

3. STEPS IN RECOMBINANT DNA TECHNOLOGY (RDT)

Step 1: Isolation of Genetic Material (DNA)
Cell wall breakdown by specific enzymes: **Lysozyme** (Bacteria), **Cellulase** (Plant cells), **Chitinase** (Fungi).
RNA removed by *Ribonuclease*; proteins removed by *Protease*.
Pure DNA precipitated as fine threads by adding **Chilled Ethanol**.

Step 2: Cutting DNA & Amplification via PCR
Cutting with restriction enzymes and isolating target fragments via Gel Electrophoresis.
Polymerase Chain Reaction (PCR): In vitro synthesis of billions of copies using 3 steps:

1. **Denaturation:** High temperature (~94°C) breaks hydrogen bonds of double stranded DNA.
2. **Annealing:** Low temperature (50–65°C) allows oligonucleotide primers to bind to complementary ends.
3. **Extension:** Thermostable **Taq Polymerase** (from *Thermus aquaticus*) extends primers at ~72°C using dNTPs.

Step 3: Ligation & Transfer into Host
Gene of interest + Vector plasmid cut with same RE → joined by **DNA Ligase** → Recombinant DNA transferred into competent host.

Step 4: Culturing Host & Bioreactors
Transformed host cultured in continuous media to obtain **Recombinant Protein** in exponential (log) growth phase.
Bioreactors (100–1000 L): Vessels providing optimal growth conditions (pH, temperature, oxygen, nutrients).

- *Simple Stirred-tank Bioreactor:* Has agitator, oxygen delivery, foam control, pH control, sampling ports.
- *Sparged Stirred-tank Bioreactor:* Sterile air bubbles sparged through system to increase surface area for oxygen transfer.

Step 5: Downstream Processing (DSP)
Series of processes including **Separation** and **Purification** of biosynthesized products.
Formulated with suitable preservatives, subjected to clinical trials & rigorous quality control tests before marketing.